SK015
Chemistry 1
Semester I
Session 2026/2027
2 Hours

SK015
Kimia 1
Semester I
Sesi 2026/2027
2 Jam

KOLEJ MATRIKULASI NEGERI SEMBILAN

UNIT KIMIA
CHEMISTRY UNIT

KIMIA PENGUKUHAN 2

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **6** questions.

Answer **all** questions.

Answer must be written in the answer booklet provided.

All steps must be shown clearly. Use a new page for each question.

Maximum marks awarded are shown in brackets at the end of each question or section.

The use of non-programmable scientific calculator is permitted.

ARAHAN KEPADA CALON

*Kertas soalan mengandungi **6** soalan*

*Jawab **semua** soalan.*

Jawapan hendaklah ditulis pada buku jawapan yang disediakan.

Semua langkah kerja hendaklah ditunjukkan dengan jelas. Gunakan muka surat baharu bagi nombor soalan yang berbeza.

Markah maksimum yang diperuntukkan ditunjukkan dalam kurungan pada hujung setiap soalan atau bahagian soalan.

Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan.

RELATIVE ATOMIC MASSES OF SELECTED

Element	Symbol	Proton number	Relative atomic mass
Aluminum	Al	13	27.0
Antimony	Sb	51	121.8
Argon	Ar	18	40.0
Arsenic	As	33	74.9
Barium	Ba	56	137.3
Beryllium	Be	4	9.0
Bismuth	Bi	83	209.0
Boron	B	5	10.8
Bromine	Br	35	79.9
Cadmium	Cd	48	112.4
Calcium	Ca	20	40.1
Carbon	C	6	12.0
Cerium	Ce	58	140.1
Cesium	Cs	55	132.9
Chlorine	Cl	17	35.5
Chromium	Cr	24	52.0
Cobalt	Co	27	58.9
Copper	Cu	29	63.6
Fluorine	F	9	19.0
Gold	Au	79	197.0
Helium	He	2	4.0
Hydrogen	H	1	1.0
Iodine	I	53	126.9
Iron	Fe	26	55.9
Krypton	Kr	36	83.8
Lead	Pb	82	207.2
Lithium	Li	3	6.9
Magnesium	Mg	12	24.3
Manganese	Mn	25	54.9
Mercury	Hg	80	200.6
Neon	Ne	10	20.2
Nickel	Ni	28	58.7
Nitrogen	N	7	14.0
Oxygen	O	8	16.0
Phosphorus	P	15	31.0
Platinum	Pt	78	195.1
Potassium	K	19	39.1
Protactinium	Pa	91	231.0
Radium	Ra	88	226.0
Radon	Rn	86	222.0
Rubidium	Rb	37	85.5
Scandium	Sc	21	45.0
Selenium	Se	34	79.0
Silicon	Si	14	28.1
Silver	Ag	47	107.9
Sodium	Na	11	23.0
Strontium	Sr	38	87.6
Sulphur	S	16	32.1
Tin	Sn	50	118.7
Tungsten	W	74	183.9
Uranium	U	92	238.0
Zinc	Zn	30	65.4

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Ionisation constant for water at 25°C *Hasil darab ion bagi air pada 25°C*	K_w	=	1.0×10^{-14} mol^2 dm^{-6}
Molar volume of gases *Isipadu molar gas*	V_m	= =	22.4 dm^3 mol^{-1} at STP 24.0 dm^3 mol^{-1} at room temperature
Speed of light in a vacuum *Laju cahaya dalam vakum*	c	=	3.0×10^8 m s^{-1}
Specific heat of water *Muatan haba tentu air*		= = =	4.18 kJ kg^{-1} K^{-1} 4.18 J g^{-1} K^{-1} 4.18 J g^{-1} °C^{-1}
Avogadro's number *Nombor Avogadro*	N_A	=	6.02×10^{23} mol^{-1}
Faraday constant *Pemalar Faraday*	F	=	96500 C mol^{-1}
Planck's constant *Pemalar Planck*	*h*	=	6.6256×10^{-34} J s
Rydberg constant *Pemalar Rydberg*	R_H	= =	1.097×10^7 m^{-1} 2.18×10^{-18} J
Molar of gas constant *Pemalar gas molar*	R	= =	8.314 J mol^{-1} K^{-1} 0.08206 L atm mol^{-1} K^{-1}
Density of water *Ketumpatan air*	ρ	=	1 g cm^{-3}
Freezing point of water *Takat beku air*		=	0.00 °C
Vapour pressure of water at 25°C *Tekanan wap air pada 25°C*	$\rho_{H\ O}$	=	23.8 torr

UNIT AND CONVERSION FACTOR

VOLUME	1 L	=	1 dm^3
Isipadu	1mL	=	1 cm^3
ENERGY	1 J	=	1 kg m^2 s^{-2} = 1 N m = 1×10^7
Tenaga	1 calorie	=	4.184 Joule
	1eV	=	1.602×10^{-19} J
PRESSURE	1 atm = 760 mmHg	=	760 torr = 101.325 kPa = 101325 N m^{-2}
Tekanan			
OTHERS	1 Faraday (F)	=	96 500 C coulomb
Lain-lain	1 Newton (N)	=	1 kg m s^{-2}

Answer **all** questions.

1. a) Silver has two naturally occurring isotopes with the following masses and natural abundances:

Isotope	Mass (amu)	Abundance (%)
Ag-107	106.9051	51.84
Ag-109	108.9048	48.16

i. Calculate average atomic mass of Ag.
ii. The silver ion, Ag^+, has 46 electrons and 61 neutrons. Write the isotopic notation of the Ag atom.

[4 marks]

b) Concentrated hydrochloric acid is an aqueous solution that is commonly used as a laboratory reagent. Calculate the mass of HCl contained in 500 mL of a solution if the density of the solution containing 37.2% of HCl by mass is 1.19 gmL^{-1}.

[4 marks]

c) Balance the following redox reaction in acidic medium:

$$SO_3^{2-} + MnO_4^- \rightarrow SO_4^{2-} + Mn^{2+}$$

[3 marks]

d) Magnesium hydroxide, $Mg(OH)_2$, is an antacid that is used to relieve indigestion, sour stomach and heartburn. The equation for the preparation of $Mg(OH)_2$ is:

$$MgCl_2(aq) + 2NaOH(aq) \rightarrow Mg(OH)_2(s) + 2NaCl(aq)$$

In an experiment, 15.1 g of $MgCl_2$ reacts with 9.35 g of NaOH.

i. Calculate the theoretical yield (in grams) of $Mg(OH)_2$.
ii. Determine the percentage yield if 6.00 g $Mg(OH)_2$ is obtained at the end of the experiment.

[10 marks]

2. a) The wavelength that produces a line A in the Paschen series is 1875 nm.

i. Calculate the frequency.
ii. Determine the electron transition that forms the line A.

[5 marks]

b) Gallium is used in light-emitting diodes (LEDs) for backlighting. The proton number of Ga is 31.

i. Write the electronic configuration of element Ga.
ii. Write a set of quantum numbers for the highest energy electron in Ga.
iii. Sketch the shape(s) of orbitals occupied by the electron(s) with the highest principal quantum number.
iv. What is the most stable ion of Ga? [5 marks]

3. a) Nitrogen trichloride, NCl_3, is formed in swimming pools when chlorine reacts with nitrogen compounds found in sweat and saliva.

i. Draw the Lewis structure of NCl_3 and determine its molecular geometry.
ii. Deduce the polarity of NCl_3.
iii. Describe the hybridisation process of the central atom and draw the overlapping of orbitals for the formation of bonds in NCl_3.

[13 marks]

b) Explain the following observations.

i. The boiling point of CBr_4 is higher than CF_4.
ii. Aluminium has a higher melting point than sodium.

[4 marks]

4. a) A mixture of 8.27 g of N_2 and 20.4 g of O_2 is transferred to a closed cylinder with a volume of 5.00 L at 25°C.

i. Determine the total pressure of gases in the cylinder.
ii. Calculate the partial pressures of N_2 and O_2 in the cylinder.

[6 marks]

b) The boiling point of propanol, $CH_3CH_2CH_2OH$ is 97°C whereas the boiling point of propanal, CH_3CH_2CHO is 49°C.

Explain the observation in terms of intermolecular forces and vapour pressure.

[3 marks]

5. a) Hydrogen bromide decomposes according to the equation below

$$2HBr(g) \rightleftharpoons H_2(g) + Br_2(g)$$

At sample of 0.600 mole HBr is introduced into an evacuated flask of volume 2.00 L and kept at an elevated temperature, T. When the system comes to equilibrium, it is found that the flask contains 0.104 mole each of H_2 and Br_2. Calculate K_c at this temperature.

[5 marks]

b) The following reaction occurs at 25 °C.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \quad \Delta H = +ve$$

Explain the shift in equilibrium position of the reaction if the following changes are applied:

i. decrease the pressure.
ii. increase in temperature.

[4 marks]

6. a) A buffer solution is prepared when 200 mL of 2.0 M NH_3 and 800 mL of 0.8 M NH_4Cl are mixed. [Given the base dissociation constant, K_b for NH_3 is 1.8×10^{-5}]

i. Calculate the pH of the prepared buffer solution.

[5 marks]

ii. Explain how the pH of this buffer solution is maintained when a small amount of strong base is added. Write a suitable equation in your explanation.

[3 marks]

b) The molar solubility of $PbCl_2$ is 1.44×10^{-2} M at a certain temperature. Calculate K_{sp} for $PbCl_2$. Determine the maximum mass (in grams) of $PbCl_2$ that will dissolve in 0.250 L of water at 25 °C.

[6 marks]

-END OF QUESTION PAPER-